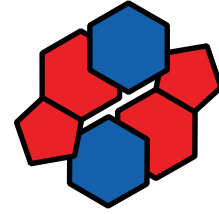




**Jordan University of
Science & Technology**



JUST | Department of
Biotechnology and
Genetic Engineering

Course Syllabus for Molecular Genetics BT341

Instructor

Dr. Osamah Batiha, Ph.D
Associate professor of cell and molecular genetics
Office: PH1-L1

Office hours:

Sundays- Wednesdays (Office): 11 am-1 pm
Thursdays: only by appointment

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Course:	Molecular genetics																				
Semester:	Second Semester																				
Course Coordinator:	Dr. Osamah Batiha																				
Lecturer:	Dr. Osamah Batiha																				
Language:	English																				
Assignment in Curriculum	Compulsory Course in 1st Semester of the third year																				
Course Units/Credit hours	Lecture: 3 credit hours																				
Students workload:	<table><tr><td></td><td>Contact hours</td><td>Private study</td></tr><tr><td>Lecture:</td><td>40</td><td>90</td></tr><tr><td>Assignment:</td><td>1</td><td>4</td></tr><tr><td>Exams & Quizzes:</td><td>4</td><td>15</td></tr><tr><td>Sum</td><td>45</td><td>109</td></tr><tr><td colspan="3">Total Sum: 154 hours</td></tr></table>				Contact hours	Private study	Lecture:	40	90	Assignment:	1	4	Exams & Quizzes:	4	15	Sum	45	109	Total Sum: 154 hours		
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Assignment:	1	4																			
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Sum	45	109																			
Total Sum: 154 hours																					
Credits	6 ECTS																				
Prerequisites according to examination regulations:	General Biology 2																				
Recommendations:	Passing the general biology course in the first academic year																				
Course Description	This course introduces you to the laws of heredity and the study of inherited traits. Topics include Mendelian and non-Mendelian inheritance, structure and function of chromosomes and genomes, biological variation resulting from recombination, mutation, and selection, and population genetics.																				
Learning outcomes:	After completing this course, students should be able to: 1- Understand the basics of inheritance rules 2-Understand chromosome and gene structure and function 3- Apply probability rules to predict the risk of reoccurrence of genetic traits and diseases. 4-Understand how genetic and physical maps were constructed 5- Understand and apply genetic tools to test/discover a genetic problem																				

Summary indicative content:	<u>Lecture</u> • CH.1 Introduction to genetics and heredity • CH.2 Mendel's principles of heredity Background: The Historical Puzzle of Inheritance, Genetic Analysis According to Mendel, Mendelian Inheritance in Humans • CH.3 Extensions to Mendel's laws Extensions to Mendel for Single-Gene Inheritance, Extensions to Mendel for Multifactorial Inheritance • CH.4 The chromosome theory of inheritance RECORDED LECTURES Chromosomes: The Carriers of Genes, Sex Chromosomes and Sex Determination, Mitosis: Cell Division That Preserves Chromosome Number, Meiosis: Cell Divisions That Halve Chromosome Number, Gametogenesis, Validation of the Chromosome Theory, Sex-Linked and Sexually Dimorphic Traits in Humans • CH.5 Linkage, recombination, and the mapping of genes on chromosomes Gene Linkage and Recombination, Recombination: A Result of Crossing-Over During Meiosis, Mapping: Locating Genes Along a Chromosome, The Chi-Square Test and Linkage Analysis, Tetrad Analysis in Fungi Mitotic Recombination and Genetic Mosaics • CH.6 DNA structure, replication, and recombination RECORDED LECTURES Experimental Evidence for DNA as the Genetic Material The Watson and Crick Double Helix Model of DNA, Genetic Information in Nucleotide Sequence, DNA Replication, Recombination at the DNA Level • CH.7 Mutations Mutations: Primary Tools of Genetic Analysis, Molecular Mechanisms of Mutation, What Mutations Tell Us About Gene Structure, What Mutations Tell Us About Gene Function • CH.8 Gene Expression The Genetic Code, Transcription: From DNA to RNA Translation: From mRNA to Protein, Differences in Gene Expression Between Prokaryotes and Eukaryotes, The Effects of Mutations on Gene Expression and Function • CH.9 Digital analysis of genomes Fragmenting DNA, Cloning DNA Fragments, Sequencing
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	DNA, Sequencing Genomes, Finding the Genes in Genomes, Bioinformatics: Information Technology and Genomes
Assessment:	The final mark of the module consists of: Written exams: - Midterm exam (30% of the final mark): composed of MCQ, short answer essay questions and/or problem solving - Final exam (40% of the final mark): composed of MCQ, short answer essay questions and/or problem solving Quizzes and assignments (30% of the final mark): quizzes are composed of MCQ. Assignments might be given based on the course needs, and it will be included among the quizzes weight. Quizzes: paper or through the e-learning
Teaching style:	<u>Lecture:</u> Lecture: On campus (Sundays and Tuesdays). Recorded lectures will be also uploaded (asynchronously) on the E-learning
Indicative Bibliography/Sources:	Genetics: from Genes to Genomes (6th Edition) by Hartwell, Hood, Goldberg, Reynolds, Silver, McGraw Hill

Rules:

- 1- Your attendance either physical or online is mandatory according to the university rules**
- 2- Make-up exams will be only given for absent students with a valid/approved reasons**
- 3- Communication between you and the instructor should be within the office hours only or by appointment**
- 4- Assignments will be worked out in groups of 5 students -max- and should be submitted written or printed. Late submitted assignments results in 10% marks' deduction per day. Copying and plagiarism are not tolerated at all and will result in Zero mark.**